



Attachment and social support in the prediction of psychopathology among young adults with and without a history of physical maltreatment^{☆,☆☆}

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Abstract

Objective: The purpose of this study was to examine the roles that social support and attachment play with regard to psychopathology among young adults with and without a history of physical maltreatment. Attachment was conceptualized in terms of the dimensions of view of self and view of other. Attachment and social support were examined individually and concurrently as protective factors.

Method: The sample consisted of 956 young adults, 294 of whom had a history of physical abuse. Individuals filled out a series of questionnaires inquiring about current attachment, social support, and psychopathology symptoms. A regression design was used, examining how well attachment and/or social support predicted current psychopathology.

Results: Results indicated that attachment security, particularly when characterized by a positive view of self, strongly predicted lower levels of psychopathology, irrespective of abuse status. Notably, view of self was a substantially larger predictor than was view of other or social support for individuals with and without a history of physical maltreatment. Among maltreatment subtypes, psychological abuse emerged as the largest predictor of psychopathology.

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Conclusions: The current study suggests that the attachment dimension of view of self is fundamental in predicting current psychopathology level, irrespective of abuse history. In addition, the subtype of psychological abuse deserves further investigation as a risk factor.

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Maltreatment as a risk factor has become a significant area of study in developmental psychopathology. Maltreated children exhibit difficulties in most developmental domains and in their ability to resolve stage-salient developmental issues successfully (Cicchetti, 1989). Maltreatment appears to have a lasting impact on individuals and is associated with increased risk of psychopathology in adulthood (Alexander et al., 1998; Muller, Goebel-Fabbri, Diamond, & Dinklage, 2000; Muller, Sicoli, & Lemieux, 2000; Roche, Runtz, & Hunter, 1999). There is, however, no consistent set of symptoms for abuse survivors, and outcomes may even include adaptive functioning (Runtz & Schallow, 1997).

The concept of resilience has been described as the capacity for successful adaptation despite challenging or threatening circumstances (Waller, 2001). Researchers have identified multiple protective factors that may promote resilience in maltreated individuals. The identification of protective factors is beneficial insofar as such information may be used to promote resilience in individuals at risk of psychopathology. Given the multiplicity of possible outcomes and the interaction of multiple risk and protective factors, research designs based on linear main effects may not be adequate (Cummings, Davies, & Campbell, 2000; Rutter, 1990). Of interest are the processes or mechanisms by which some individuals are able to overcome adversity (Cummings et al., 2000). Researchers have more recently employed more complex research designs to delineate how protective factors may interact and exert their effects (Muller, 1996; Muller & Lemieux, 2000; Muller, Sicoli, et al., 2000; Roche et al., 1999). In the current study, two such protective factors (attachment and social support) will be examined in greater detail.

Attachment theory and research

Bowlby (1980a, 1980b, 1982) emphasized the role of child-caregiver interactions in the development of cognitive, behavioral, and affective systems. He defined attachment as an emotional bond characterized by proximity-maintaining behavior with a specific person, especially under stressful circumstances. Through interactions with primary caregivers, children develop internal working models of themselves, others, and relationships (Bowlby, 1980a). A securely attached child, who has generally experienced consistent and sensitive care, is likely to develop a model of the other person as responsive, themselves as worthy of love and competent in eliciting attention, and the relationship as rewarding. An insecurely attached child, who has generally not experienced consistent or sensitive care, will likely develop a model of other people as unpredictable, themselves as unworthy of love and unable to gain positive attention, and the relationship as unsatisfactory (Cicchetti, Toth, & Lynch, 1995). Working models combine to form an organizational framework for the child's use in future social situations, influencing his/her perceptions, expectations, and behaviors (Cassidy & Mohr, 2001; Cicchetti et al., 1995; Sroufe, Carlson, Levy, & Egeland, 1999; Wekerle & Wolfe, 1998).

Attachment researchers consider the attachment system to be active throughout the life span (Bowlby, 1980a; Collins & Read, 1990, 1994; Mikulincer, Gillath, & Shaver, 2002). Adult working models are not developed solely from the parent-child relationship, but also from relationships with peers and important others (Collins & Read, 1994). The development of multiple working models from different relationship experiences provides adults with the flexibility they need to function adaptively. Researchers have found that attachment styles are not entirely static, but may change with significant life events or new relationship experiences (Cozzarelli, Karafa, Collins, & Tagler, 2003; Weinfield, Whaley, & Egeland, 2004).

Bartholomew and Horowitz (1991) have conceptualized adult attachment based on internal working models of the self and others. An individual may hold a “self” model as either positive (self as worthy of love) or negative (self as unworthy), and an “other” model as either positive (other as responsive and loving) or negative (other as unresponsive and not loving). Different combinations of the two poles of self and other models produce four attachment categories: secure (positive-positive), preoccupied (negative-positive), fearful (negative-negative), and dismissing (positive-negative). Secure individuals view themselves and others as positive and worthy of love, and find relationships rewarding. Preoccupied individuals carry a sense of unworthiness, and strive for self-acceptance by gaining acceptance from valued others. Fearful individuals view themselves as unworthy, and others as rejecting and untrustworthy. They tend to avoid close involvement with others in order to avoid rejection and hurt. Dismissing individuals consider themselves to be worthy, but they dismiss relationships in order to escape disappointment and to maintain a sense of independence and invulnerability. For further information on adult attachment dimensions, see Muller, Kraftcheck, and McLewin (2004).

An advantage to considering both one’s model of self and other is the resulting ability to determine their relative contributions to the development of psychopathology or resilience. Researchers acknowledge the importance of both, yet attachment patterns characterized by different underlying models show varying patterns of adaptation and risk (Keelan, Dion, & Dion, 1998; Muller, Sicoli, et al., 2000). An understanding of how the self and other poles contribute to psychopathology may help delineate how some high risk individuals appear resilient and how others develop psychopathology (Muller & Lemieux, 2000).

Research with formerly maltreated individuals has found that insecure attachment styles, particularly those characterized by a negative view of self, may place such individuals at high risk for the development of psychopathology. Research on adult survivors of childhood abuse (Alexander et al., 1998; Muller & Lemieux, 2000) has found greater psychopathology associated with those individuals holding preoccupied or fearful (negative view of self) attachment patterns. Similarly, Muller, Goebel-Fabbri, et al. (2000a) found that among adults maltreated as children, a negative view of self was a better predictor of posttraumatic stress symptoms than was a negative view of others. Roche et al. (1999), in a study of women with a history of childhood sexual abuse, found that one’s view of self appeared to be the most important attachment dimension for predicting adjustment.

It is important to determine by what processes attachment security, and particularly a positive model of self, exert their apparent protective effects among adult survivors of abuse. One possible link is that of success in adult relationships. A considerable body of research exists on the relation between attachment and various aspects of intimate relationships. Secure attachment has been associated with both less conflict (Wekerle & Wolfe, 1998) and better quality (Cohn, Silver, Cowan, Cowan, & Pearson, 1992; Feeney, Noller, & Callan, 1994) within close relationships. The communication styles of securely attached individuals in close relationships have also been noted as more open, productive, and flexible than that of insecurely attached adults (Collins & Read, 1990; Keelan et al., 1998). Since attachment is so closely linked to intimate relationships in adulthood (Hazen & Shaver, 1987), and these close relation-

ships are potentially important sources of social support in times of stress (Johnson & Williams-Keeler, 1998; Whiffen & Oliver, 2004), the constructs of social support and attachment need to be examined concurrently.

Protective effects of social support

Social support has received empirical attention as a protective factor among individuals at risk (Schumm, Vranceanu, & Hobfoll, 2004). Researchers suggest that social support may function by either buffering an individual from life stress (Brewin, Andrews, & Valentine, 2000), or by directly enhancing an individual's well-being (Schumm et al., 2004). Longitudinal studies of high risk samples identify a supportive relationship with an available adult as common among those who are functioning well compared to their non-resilient peers (Egeland, 1997; Egeland, Carlson, & Sroufe, 1993; Werner, 1993).

Clinicians have suggested that the marital relationship can serve as a “recovery environment” for those coping with the aftermath of trauma (Johnson, 2004; Johnson & Williams-Keeler, 1998), providing a relationship experience that contradicts the learning that occurred during the trauma. Research with adult survivors of physical and sexual abuse also attests to the importance of social support for resilience. A meta-analysis of 11 studies correlated a lack of social support with PTSD symptoms among trauma survivors (Brewin et al., 2000). Runtz and Schallow (1997) examined social support among 302 adults who had experienced physical or sexual abuse. With regard to adaptive functioning, they found that 55% of the variance was attributed to social support. A significant proportion of the variance remained unexplained by this model, leading the authors to suggest the investigation of factors influencing social support.

Social support and attachment

Researchers and theorists are beginning to investigate how some people are able to benefit from social support, while others are not. Sarason, Pierce, and Sarason (1990) view “a sense of social support” as a personality characteristic arising from early relationship experiences. According to Sarason et al. (1990), the attachment process in childhood is of primary importance, enabling children to develop expectations of others' availability, and develop a sense of self-efficacy in relationships. A secure attachment experience in infancy and childhood lays the foundation for an individual's ability to establish, tolerate, and maintain intimacy in relationships. Such individuals are likely to seek emotional support from close relationships when under stress, perceive a high level of available support, and be satisfied with the results. Individuals characterized by a sense of social support are thus expected to be more effective copers than those without this personality characteristic. Ognibene and Collins (1998) found that securely attached individuals were more likely to seek social support in times of stress than those with insecure attachment patterns. They did not, however, investigate any benefits stemming from such support.

Recent studies have successfully employed “reciprocal effects” models of developmental psychopathology (Muller, 1996; Muller & Lemieux, 2000). The concept of reciprocal effects stems from the assertion that individuals are active participants in their environment, bringing to new situations their

beliefs and experiences. The individual is viewed as actively involved in the developmental process by bringing competencies to each phase of development, and by behaving in ways that elicit environmental responses supporting prior adaptation (Cicchetti et al., 1995; Cummings et al., 2000; Waller, 2001). A reciprocal effects model of attachment, social support, and psychopathology would suggest that securely attached individuals would derive more benefit from socially supportive relationships available to them. As such, there should be a synergistic relationship (a positive interaction) between attachment and social support in the promotion of functioning among at-risk adults.

Muller and Lemieux (2000) empirically investigated the interaction between social support and attachment in promoting adaptive functioning among formerly maltreated adults. The authors found a significant positive correlation between lack of social support and increased psychopathology, and between a negative view of self and increased psychopathology. Further, the relationship between a negative view of self and psychopathology was stronger for those low on social support than for those high on social support. A negative view of others was not significantly related to psychopathology. These results are consistent with other studies that demonstrate the beneficial effects of social support. They also show differential effects according to attachment dimension, and are suggestive of an interaction between social support and the attachment dimension of view of self in predicting psychopathology.

Gaps in previous research

As discussed above, research on the interaction between social support and attachment in the prediction of psychopathology has demonstrated some compelling findings. Nevertheless, there remain methodological and theoretical gaps that limit prior research. First, previous studies on this topic have tended to rely almost exclusively on small samples (e.g., Muller & Lemieux, 2000; Muller, Sicoli, et al., 2000; Ognibene & Collins, 1998; Roche et al., 1999), which can increase sampling error and reduce power thereby obscuring potentially significant effects. This is particularly problematic in the case of Muller and Lemieux (2000), where the synergistic effects of attachment and social support were not consistently found. A significant interaction was found between attachment and social support using an ANOVA model, with social support divided into high and low levels using a median split. However, no such synergistic effects were found in multiple regression analyses. As is often the case in small sample research, this inconsistency in results may have been an artifact of sampling error due to the relatively small sample size used ($N = 66$). When examining the interaction of social support and attachment in the prediction of psychopathology using a regression model, a much larger sample size would allow for a substantially stronger test of the proposed synergistic effects.

Second, previous studies have examined social support and/or attachment constructs almost exclusively in samples of maltreated individuals, failing to include a control group (Alexander et al., 1998; Muller & Lemieux, 2000; Muller, Lemieux, & Sicoli, 2001; Runtz & Schallow, 1997). While social support and attachment security have emerged as significant protective factors in at-risk samples, it remains unclear how important these factors are in individuals who have not experienced significant abuse. People who have not experienced significant trauma in their life may not need to rely on the same resources as those with a trauma history in order to function adaptively. The inclusion of a control group would allow for a comparison of how strongly attachment and social support predict psychopathology in those with abuse histories versus those who have not experienced maltreatment, speaking to the robustness of these protective factors in different samples, and in the population at large.

The current study

The purpose of this study was to examine further attachment, social support, and the relationship between these two constructs in the prediction of psychopathology among young adult survivors of physical maltreatment, and among young adults without a history of physical maltreatment. Specifically, we examined the main effects of attachment security (view of self and view of others) as well as social support in predicting lower levels of psychopathology. In order to expand upon prior research, we also examined the potential synergistic relationship (positive interaction) between attachment and social support with regard to their role in predicting lower levels of psychopathology. Note that we conceptualized attachment as consisting of self and other models. The interactions of attachment security and social support, and view of self and social support were tested. View of other has not been found to vary with social support level in predicting psychopathology (Muller & Lemieux, 2000). Note also that in order to minimize the effects of sampling error, a very large sample size was utilized in this study.

It was hypothesized that in the prediction of lower levels of psychopathology: (1) individuals with higher attachment security and a more positive view of self and others would function better; (2) view of self would emerge as a very strong predictor of better functioning; (3) view of self, in comparison with view of other, would emerge as the larger predictor; (4) individuals reporting more social support would report better functioning; (5) social support and attachment would positively interact in the prediction of better functioning (synergistic effect).

Method

Participants

Participants consisted of undergraduate students at an urban Canadian university. Out of 1,003 students who agreed to participate, 956 students completed questionnaire packages in their entirety. A demographic questionnaire was administered to participants, which provided information regarding age, gender, year at university, cultural background, and religion. In addition, socioeconomic status was derived from questions regarding parental income, occupation, and education (Hollingshead Index; Mueller & Parcel, 1981).

Demographic characteristics. The sample consisted of 82.4% female participants. The majority of participants were between the ages of 16 and 24, with 42.7% between 16 and 19 years old, and 48.6% aged 20–24 years. A smaller percentage of participants identified themselves as being 25 years or older (8.7%). The majority of participants were in their first year of university (54.5%). Participants' ethnicity was predominantly Euro-Canadian (58.1%), followed by Asian-Canadian (14%), other (11.6%), South Asian-Canadian (8.7%), and African/Caribbean-Canadian (7.6%).

Predictor measures

Relationship Scales Questionnaire (RSQ; Griffin & Bartholomew, 1994a). The RSQ is a 30-item questionnaire designed to measure a four-category model of adult attachment. Individuals rate each item on a 5-point Likert scale (1 = “not at all like me” to 5 = “very much like me”) according to how well each item describes their attachment category. These include: (a) secure attachment (e.g., “I find it easy to get

emotionally close to others”), (b) fearful attachment (e.g., “I worry that I will be hurt if I allow myself to become too close to others”), (c) preoccupied attachment (e.g., “I worry that others don’t value me as much as I value them”), and (d) dismissing attachment (e.g., “I prefer not to have other people depend on me”). The mean rating for each of the four subscales is computed, creating four continuous variables corresponding to attachment styles.

Each attachment style is characterized by a particular underlying model of self and other (Schafer & Bartholomew, 1994), which can be scored as two separate continuous variables. Construct validity of the self and other dimensions has been demonstrated (Bartholomew, 1990). In addition, convergent and discriminant validity of these two dimensions have been demonstrated (Griffin & Bartholomew, 1994b). Moderate to high test-retest reliability and stability coefficients were demonstrated over an 8-month period, ranging from $r = .81$ to $.84$ for view of self, and from $.72$ to $.85$ for view of other (Schafer & Bartholomew, 1994).

Relationship Questionnaire (RQ; Bartholomew & Horowitz, 1991). The RQ is an adaptation of Hazen and Shaver’s (1987) three-category adult attachment measure. The RQ is designed to test Bartholomew’s (1990) four-category model of attachment. It consists of four short paragraphs describing four attachment styles: secure, fearful, preoccupied, and dismissing. In the first part, individuals choose the paragraph that best describes them in their closest relationships. In the second part, individuals rate each description according to personal applicability on a 5-point scale, ranging from 1 (not at all like me) to 5 (very much like me). As with the RSQ, four continuous variables corresponding to attachment styles result, and self and other models can be calculated. This scale may be combined with the RSQ by changing attachment prototype scores to z-scores, and then averaging parallel scores (Bartholomew, 2001, personal communication; Ognibene & Collins, 1998). The resulting attachment prototype scores can be used to calculate the self and other model dimensions. Examinations of the RQ have demonstrated both discriminant and convergent validity (Bartholomew & Horowitz, 1991; Griffin & Bartholomew, 1994a).

Multi-Dimensional Support Scale (MDSS; Neuling & Winefield, 1988; Winefield, Winefield, & Tiggemann, 1992). The MDSS is a self-report measure of social support where individuals rate the amount and adequacy of supportive behaviors received from three sources. The MDSS consists of 19 items per source describing actions expressing attentiveness, empathy, distraction, attachment, instrumental help, direct information, and modeling. The current study used as sources: “family and close friends;” “other people of about your age that you know that are like you;” and “people in some sort of authority over you.” Individuals rate how often each type of supportive behavior was received within the past month from each source on a 4-point scale ranging from 1 (never) to 4 (usually/always). Each item is also rated for satisfaction on a 3-point scale, according to whether they would have liked the behavior 1 (more often), 2 (less often), or 3 (it was just right).

Research using the MDSS has demonstrated high internal reliability when administered to a sample of young adults (Winefield et al., 1992) and modest reliability with cancer patients (Neuling & Winefield, 1988), with Cronbach’s alphas ranging from $.81$ to $.90$ in the former study, and from $.50$ to $.81$ in the latter study. The scale has also shown predictive validity for adjustment and measures of psychological well-being, including self-esteem, depression, anxiety, and health (Neuling & Winefield, 1988; Winefield et al., 1992). Winefield et al. (1992) recommended that items concerning distraction as a supportive behavior be eliminated from the MDSS in future research, as it was not rated as helpful by their sample. Thus, the current study excluded the three distraction items from analyses.

Record of Maltreatment Experiences, Self-Report (ROME; Wolfe & McGee, 1994). The ROME Self-Report is a retrospective measure of victimization occurring from birth to 17 years. It is comprised of

five subscales designed to assess the type and degree of maltreatment an individual has experienced or witnessed. The subscales include: (1) positive parenting practices, (2) psychological maltreatment, (3) physical maltreatment, (4) exposure to family violence, and (5) sexual abuse (Wolfe & McGee, 1994). Cronbach's alpha reliabilities for each of the subscales are .82, .89, .81, .81, and .91, respectively (McGee, 1995, personal communication). Test-retest reliability coefficients range from .70 for psychological maltreatment to .93 for sexual abuse (Wolfe & McGee, 1994).

Physical abuse items are categorized as major or minor physical abuse experiences, consistent with past categorization criteria (Muller, 1996). Four physical abuse items thought to represent less potentially damaging contact experiences are classified as minor (i.e., "threw or smashed or hit or kicked something;" "threw something at you;" "pushed, grabbed, or shoved you;" "slapped you"). Five physical abuse items reflecting potentially damaging or lethal contact experiences are classified as major (i.e., "kicked, bit or hit you with a fist;" "hit or tried to hit you with something;" "beat you up;" "threatened you with a knife or gun;" "used a knife or gun"). Individuals rate the frequency with which each statement occurred with reference to both their primary male and female caregiver according to the following scale: 0 = never; 1 = very infrequently (once or twice during childhood); 2 = occasionally (1–3 times per year); 3 = frequently (1–2 times per month); and 4 = very frequently (once a week or more). Domestic physical violence items were the same as the above physical abuse items, occurring from mother to father and father to mother.

The psychological maltreatment subscale consisted of nine items, which are potentially psychologically damaging (i.e., "ridiculed or made fun of you in front of others;" "insulted you, put you down, or called you names;" "told you that you were a burden, were unwanted"). Individuals rate the frequency with which each statement occurred with reference to both their primary male and female caregivers according to the same scale as the physical abuse items.

Finally, sexual abuse history was measured by recording individuals' sexual experiences prior to the age of 14 with someone at least 5 years older, or with someone around the same age when the experience involved force. The experiences presented ranged from "an invitation or request to do something sexual" to "anal intercourse." Individuals responded with either "yes" or "no" to each statement.

Confirmatory factor analyses indicated that while abuse dimensions of physical, psychological, and domestic violence ran parallel to each other in forming a higher order abuse construct, sexual abuse appeared to run separately. Psychometric reasons responsible for this observation were unclear, thus sexual abuse was not included as a variable in further analyses.

Outcome measures

Young Adult Self-Report (YASR; Achenbach, 1997). The YASR is a self-report questionnaire that includes 110 items measuring emotional and behavioral problems and 14 items reflecting socially desirable traits. Participants are asked to rate each item as it describes them over the past six months ("0 = not true," "1 = somewhat or sometimes true," or "2 = very true or often true"). The measure allows for the scoring of individual problem subscales as well as "internalizing," "externalizing," and total psychopathology scales. Cronbach's alpha reliability is high ($r = .96$) for the total scale (Achenbach, 1997). For an earlier and nearly identical version of the YASR, test-retest reliability was reported as .89 (Ferdinand, Verhulst, & Wiznitzer, 1995). The YASR is reported to have strong validity (Achenbach, 1997; Aronen & Soininen, 2000).

Trauma Symptom Checklist-40 (TSC-40; Briere & Runtz, 1989; Elliot & Briere, 1992). The TSC-40 is a 40-item self-report questionnaire that evaluates adult symptomatology arising from childhood or adult traumatic experiences. It consists of six subscales (anxiety, depression, dissociation, postsexual abuse trauma, sleep disturbance, and sexual problems) and a total score. Individuals rate how often each item has occurred in the last 2 months on a 4-point scale from “never” to “often.” Reliability of the TSC-40 is modest to high, with alphas ranging from .62 to .77 for the subscales, and .90 for the total score (Elliot & Briere, 1992). The scale has also demonstrated validity with regard to traumatic experiences, such as childhood abuse (Gold, Milan, Mayall, & Johnson, 1994; Wind & Silvern, 1992).

Procedure

This project was approved by the York University research review committee. The study involved two waves of data collection. The current study examines the first wave of data, which was collected between Fall 1999 and Winter 2001. The majority of the participants was enrolled in an Introductory Psychology course (59.3%) and received one bonus mark towards their final grade for their participation. The remaining participants (40.7%) were students taking second and third year psychology courses, who either received one bonus mark, or had the opportunity to participate in a draw for prizes (i.e., movie certificates) in return for completing the study. Participants signed up for the study based on a description of the research as focusing on “important relationships in an individual’s environment.” Participants individually completed a series of questionnaires in groups of 15–20 people. The questionnaire packet consisted of the measures outlined in the measures section. There were six orderings of the measures, always beginning with demographic information and ending with the Record of Maltreatment Experiences (ROME). Analyses of variance conducted on each of the outcome measures indicated no significant order effects.

Allocation of participants to groups. In the current sample, 294 participants met criteria for having experienced some form of physical maltreatment according to their reports on the Record of Maltreatment Experiences (see Measures section). Criteria employed to determine abuse status are based on empirical research indicating that a very severe act impacts even at lower frequencies of occurrence, while frequency is a more important factor with regard to impact for less severe experiences (Manly, Cicchetti, & Barnett, 1994). Thus, in the current study individuals were considered physically maltreated if they had experienced severe acts of abuse at low frequencies, had experienced a high frequency of more minor abusive acts, or had experienced some combination of major and minor abusive acts.

The minimum maltreatment criteria consisted of the endorsement of one of the following experiences: At least one minor physical abuse experience with frequencies ranging from “1–2 times a month” to “once a week or more;” at least two major physical abuse experiences with a frequency of “once or twice during childhood” plus at least one minor physical abuse experience of any frequency; at least one major physical abuse experience with a frequency of “1–3 times a year” plus at least one minor physical abuse experience of any frequency. Twenty-two percent of the abuse sample experienced at least one major physical abuse item at the highest frequencies (“once a week or more”). Five-hundred and seventy-five participants not meeting the above criteria comprised a non-physical abuse group. Participants reporting sexual abuse experiences were also excluded from this group.

Statistical power analysis

A statistical power analysis was calculated in order to determine adequacy of sample size to detect hypothesized effects. Green's (1991) guidelines for regression power analysis were followed. With an alpha level of .05, a medium effect size assumed, and seven predictor variables (the maximum for this study), 106 participants would be needed to detect multiple regression effects. Further, 111 participants would be needed to detect individual predictor contributions at the same power level. Thus, the analysis confirmed that the smallest sample analyzed in this study (294 physically maltreated individuals) should provide more than adequate power for the proposed effects to be detectable.

Statistical model

A regression design was employed in order to determine the relative predictive power of attachment and social support in predicting lower levels of psychopathology. Standard multiple regression analyses were performed separately on the physical abuse group, non-physical abuse group, and entire sample. It was deemed important to examine hypotheses separately in abused and non-abused individuals to ascertain whether protective factors operated differently in these two populations. Previous research had examined these questions in abuse samples only. Abuse subtypes were included as predictor variables in regressions performed on the entire sample, in order to ascertain the relative predictive power of risk (maltreatment) and protective (attachment and social support) factors. Two regressions were performed for each sample, one with the TSC-40 as the criterion variable, and one with the YASR as the criterion variable. Each regression included social support, view of self, view of other, and the interaction of social support and view of self as predictor variables. For each regression analysis, all predictor variables were entered simultaneously. All of the multiple regression analyses were significant, and thus individual predictors were examined. Results are discussed in terms of the standardized beta weights.

A second set of regression analyses was run with social support, secure attachment, and the interaction of social support and secure attachment as predictor variables. The results of that analysis were very similar to the results presented here (where attachment was broken down into the self and other poles), and may be obtained from the authors on request. They were excluded from this report for the sake of brevity.

Pearson correlations and multiple regression analyses were conducted using a significance level of .05. As the large sample size and high power were likely to produce significant results with sometimes small effects, correlations and beta weights were interpreted based on small ($r = .10$), medium ($r = .30$), and large ($r = .50$) effect sizes, according to Cohen's (1988) criteria. Note that throughout the results section, psychopathology measures were reverse scored such that higher scores indicate better functioning.

Results

Cronbach's alpha coefficients were generated for the scales used in this study, yielding high reliabilities ranging from .86 to .95. In order to determine whether demographic variables should be included in the analyses, two one-way ANOVAs were conducted for each demographic variable, including age,

Table 1

Correlations among maltreatment, attachment, social support, and psychopathology measures ($N = 956$)

Variables	1	2	3	4	5	6	7	8	9
1. Psychological abuse	–								
2. Physical abuse	.64*	–							
3. Domestic violence	.32*	.46*	–						
4. Secure attachment	–.24*	–.16*	–.06	–					
5. Self model	–.27*	–.15*	–.07	^a	–				
6. Other model	–.15*	–.14*	–.05	^a	.17*	–			
7. Social support	–.29*	–.21*	–.17*	.33*	.30*	.25*	–		
8. TSC-40	–.31*	–.22*	–.16*	.44*	.52*	.24*	.29*	–	
9. YASR	–.41*	–.27*	–.18*	.49*	.55*	.26*	.31*	.76*	–

Note. TSC-40, Trauma Symptom Checklist-40; YASR, Young Adult Self Report. The TSC-40 and YASR were coded such that higher scores indicate better functioning. Self model and other model refer to positive views of self and other.

^a Self and other model dimensions form the secure attachment scale. Thus, correlations between self and other models and secure attachment were not included.

* $p < .05$.

gender, year at university, culture, religion, and socioeconomic status. The YASR and TSC-40 served as dependent measures. Results indicated that none of the demographic variables explained more than 3% of the variance, and thus they were excluded from further analyses.

Correlation analyses

Correlations were calculated in order to examine the relationships among maltreatment constructs, attachment, social support, and psychopathology measures (see Table 1). A Bonferroni correction was calculated in order to correct for familywise Type I error. As expected, protective factors (attachment constructs and social support) were positively associated with one another, and risk factors (maltreatment constructs) were positively correlated with one another. Protective factors were negatively associated with risk factors. Finally, variables were correlated in the expected direction with psychopathology and PTSD symptomatology; maltreatment subtypes were associated with higher levels of psychopathology, and protective factors were associated with lower levels of psychopathology.

Regression analyses

Physical abuse group. Table 2 displays the results of regressions conducted with the physical abuse group. Social support, view of self, and view of other were all significant predictors of psychopathology (reverse scored) as reported on both measures, with higher levels of social support, a positive view of self, and a positive view of other predicting better functioning. In both regressions, view of self emerged as the strongest predictor ($p < .001$), consistent with the hypothesis that a positive view of self, in comparison with a positive view of other, would emerge as the largest predictor of functioning. Once attachment dimensions were accounted for, social support was a modest, but statistically significant predictor, with beta weights of .12 (TSC-40; $p < .05$) and .14 (YASR; $p < .01$). The interaction of social support and view of self was not significant in either regression, suggesting that once the main effects of social support and view of self were accounted for, there was no interaction of social support by view of self.

Table 2

View of self, view of other, and social support as predictors of psychopathology in two multiple regressions (physical abuse group, $n = 294$)

Variable	TSC-40			YASR		
	<i>B</i>	<i>SE B</i>	β	<i>B</i>	<i>SE B</i>	β
Step 1						
Social support	.10	.05	.12*	.07	.03	.14**
View of self	.10	.01	.47***	.06	.01	.52***
View of other	.05	.01	.20***	.02	.01	.17***
Social support $\times$ view of self	-.01	.02	-.03	.01	.01	.02

Note. $R^2 = .35$ for TSC-40; $R^2 = .40$ for YASR ($p < .001$). TSC-40, Trauma Symptom Checklist-40; YASR, Young Adult Self Report. The TSC-40 and YASR were coded such that higher scores indicate better functioning. The view of self and view of other dimensions were coded such that higher scores represent better views of self and of other.

* $p < .05$.

** $p < .01$.

*** $p < .001$.

Non-physical abuse group. Findings from regression analyses with the non-physical abuse group are presented in Table 3. Results were similar to those found for the physical abuse group. View of self and view of other significantly predicted psychopathology (reverse scored) for both dependent measures. As expected, positive models of self and of other predicted better functioning. The self model was the strongest predictor ($p < .001$). Social support emerged as a significant predictor in one regression. Social support was a small, but significant predictor of the YASR measure ($p < .01$), but not the TSC-40. This indicates that once the self and other poles of attachment were considered, social support made a relatively small contribution in predicting psychopathology. Finally, consistent with the physical abuse group findings, the interaction of social support and model of self did not significantly add to the prediction of functioning.

Table 3

View of self, view of other, and social support as predictors of psychopathology in two multiple regressions (non-physical abuse group, $n = 575$)

Variable	TSC-40			YASR		
	<i>B</i>	<i>SE B</i>	β	<i>B</i>	<i>SE B</i>	β
Step 1						
Social support	.06	.03	.07	.05	.02	.10*
View of self	.09	.01	.46**	.05	.00	.47**
View of other	.02	.01	.20*	.02	.00	.16**
Social support $\times$ view of self	-.01	.01	-.03	.01	.01	.03

Note. $R^2 = .26$ for TSC-40; $R^2 = .31$ for YASR ($p < .001$). TSC-40, Trauma Symptom Checklist-40; YASR, Young Adult Self Report. The TSC-40 and YASR were coded such that higher scores indicate better functioning. The view of self and view of other dimensions were coded such that higher scores represent better views of self and of other.

* $p < .01$.

** $p < .001$.

Table 4

View of self, view of other, social support, and maltreatment as predictors of psychopathology in two multiple regressions (whole sample, $N = 956$)

Variable	TSC-40			YASR		
	<i>B</i>	<i>SE B</i>	β	<i>B</i>	<i>SE B</i>	β
Step 1						
Psychological abuse	−.10	.03	−.13***	−.11	.02	−.25***
Physical abuse	−.00	.04	−.00	.01	.02	.01
Domestic violence	−.08	.04	−.07*	−.03	.02	−.05
Social support	.06	.02	.08**	.03	.01	.07*
View of self	.09	.01	.43***	.05	.00	.44***
View of other	.03	.01	.13***	.02	.00	.13***
Social support $\times$ view of self	−.01	.01	−.03	.01	.01	.01

Note. $R^2 = .33$ for TSC-40; $R^2 = .40$ for YASR ($p < .001$). TSC-40, Trauma Symptom Checklist-40; YASR, Young Adult Self Report. The TSC-40 and YASR were coded such that higher scores indicate better functioning. The view of self and view of other dimensions were coded such that higher scores represent better views of self and of other.

* $p < .05$.

** $p < .01$.

*** $p < .001$.

Whole sample. Regression analyses conducted on the entire sample included psychological abuse, physical abuse, and exposure to domestic violence as additional predictors. Results are shown in Table 4. Social support emerged as a small, but significant predictor of psychopathology (reverse scored) in both regressions (TSC-40: $p < .01$; YASR: $p < .05$). The attachment constructs of view of self and view of other both emerged as significant predictors ($p < .001$), with positive view of self and view of other predicting better functioning. View of self was the strongest predictor. Finally, as in previous regressions, the interaction of social support and view of self was not significant, indicating that once individual predictors were accounted for, the interaction of view of self and social support did not contribute further in predicting functioning. Psychological abuse was a significant predictor in both regressions ($p < .001$), with higher levels of psychological abuse predicting more psychopathology. Physical abuse did not predict psychopathology in either regression. Finally, exposure to domestic violence was a small, but significant predictor of trauma symptomatology ($p < .05$), but not general psychopathology.

Discussion

The purpose of the current study was to investigate protective factors and processes in a large sample of young adults with and without a history of physical maltreatment. Specifically, attachment and social support were examined both individually as protective factors, and together as possible interactive factors in the prediction of psychopathology. The construct of attachment was further subdivided into the underlying dimensions of self and other, in order to delineate which aspect of attachment was more predictive of functioning. In all analyses, greater attachment security, particularly when characterized by a positive view of self, emerged as the strongest predictor of lower levels of psychopathology, irrespective of abuse status. View of self was a stronger predictor than either view of other or social support, consistent with prior research with abuse samples. Thus, view of self seems to be the aspect of attachment most important

to functioning in both normative and at-risk samples. The interaction of attachment and social support did not contribute to the prediction of psychopathology once the individual variables were accounted for. Finally, psychological maltreatment emerged as the largest predictor among abuse subtypes when examined with the entire sample.

Protective effects of attachment

As hypothesized, security of attachment was significantly related to, and a significant predictor of functioning. Irrespective of abuse status, individuals with greater attachment security reported lower levels of psychopathology. This is consistent with attachment theory, which proposes that attachment security acts as an “inner resource” that promotes adaptive coping (Bowlby, 1988; Mikulincer & Florian, 1998). Individuals with a history of having their needs met develop constructive and optimistic views of life problems, and thus experience less distress (Mikulincer, 1998; Mikulincer, Shaver, & Pereg, 2003). Researchers have found that securely attached individuals have a more coherent and positive self view (Florian, Mikulincer, & Bucholtz, 1995; Mikulincer, 1998), report more reciprocity and better quality in relationships (Feeney et al., 1994; Keelan et al., 1998), and exhibit less psychopathology (Bifulco, Moran, Ball, & Lillie, 2002) than insecurely attached individuals. Attachment insecurity may obstruct the development of inner resources necessary for coping with stress, thus limiting the individual’s ability to navigate successfully challenges that arise.

View of self and view of other

View of self emerged as the largest predictor of functioning across measures of psychopathology for both physically abused and non-physically abused samples. Prior research with formerly maltreated adults has also found view of self to be a critical predictor of psychopathology or adaptation (Muller & Lemieux, 2000; Muller et al., 2001). Crittenden (1997) suggested that those with negative self models may be highly vulnerable to distress. That is, these individuals appear to rely on affect rather than cognition to guide their behavior, but they have difficulty experiencing negative affects over time without resorting to maladaptive coping responses such as avoidance (Briere, 1996; Herman, 1997). Research investigating emotional biases has found that individuals holding attachment styles characterized by a negative self model tend to feel easily ashamed (fearful attachment) and disgusted (preoccupied attachment) (Magai, Hunziker, Mesias, & Culver, 2000). Further, anxiety and depressed affect are key personality traits for these individuals, increasing the impact of stress and an individual’s vulnerability to psychopathology.

Research has found that those with a negative view of self have a greater number of dysfunctional beliefs (Roberts, Gotlib, & Kassel, 1996). They may see themselves as unworthy of love and fulfillment of their needs by others, and believe that their ability to keep themselves safe from harm is impaired. Such individuals tend to blame themselves for negative events in their lives, and fail to take responsibility for positive experiences. A negative view of self can also adversely affect interpersonal functioning by hampering the ability to develop and sustain satisfying relationships (Ognibene & Collins, 1998). Such individuals may not reach out to others, and thus miss out on potential corrective experiences that can improve their sense of self in relation to others (Ognibene & Collins, 1998; Roberts et al., 1996).

As a negative self model may place individuals at increased risk of maladaptive functioning, a positive model of self may be protective. In the current study, a more positive view of self was related to

better functioning. Prior research has further found a positive self model to be related to increased optimism, better emotional health, more independence, increased resourcefulness, and less use of immature defense mechanisms (Diehl, Elnick, Bourbeau, & Labouvie-Vief, 1998). This study further confirms the importance of the self model for psychological functioning.

While both view of self and view of other significantly predicted functioning in the current study, view of self emerged as a substantially stronger predictor than did view of other. This finding was consistent across measures and groups. This finding is also consistent with prior research with formerly maltreated adults, which has found a negative view of self to be more predictive than a negative view of other with regard to both psychopathology and PTSD symptomatology (Mikulincer, Florian, & Weller, 1993; Muller et al., 2001; Muller, Sicoli, et al., 2000; Roche et al., 1999). Prior research examining the view of other dimension of attachment has found mixed results with regard to functioning. While view of other has been associated with greater difficulty in interpersonal functioning in some studies (Bartholomew & Horowitz, 1991; Griffin & Bartholomew, 1994b), research with formerly maltreated adults has not found a strong association between view of other and psychopathology (Mikulincer et al., 1993; Muller et al., 2001; Muller, Sicoli, et al., 2000; Roche et al., 1999).

Individuals holding a negative view of other may not be at risk in the same way as those holding a negative view of self. That is, individuals who hold a negative view of others may still be able to develop adaptive coping strategies (Muller et al., 2001). Such individuals may discount the importance of relationships, and instead value the importance of self-reliance (Alexander et al., 1998). While self-isolation can be viewed as a maladaptive coping strategy, as it decreases the likelihood that working models will be revised with new interpersonal experiences, it may increase one's ability to cope with life stressors by strengthening an individual's sense of self-worth. Those individuals who are able to maintain a sense of self-efficacy and self-control are more likely able to view the world as manageable, and function adaptively (Herman, 1997).

An alternative explanation for the weaker association between psychopathology and view of other is that these individuals may be underreporting inner distress. Magai et al. (2000) found that dismissing individuals, while denying anxiety, were experiencing inner conflict and distress as measured by projective tests. Ognibene and Collins (1998) suggested that avoidant individuals (fearful and dismissing) may respond to stressors by distancing themselves. They found partial support for this position, although psychopathology symptoms were not measured. Future research may benefit by employing alternative measures to self-reports in order to clarify further the degree to which those with a negative other model are at risk.

Social support and attachment

As hypothesized, social support emerged as a significant predictor of psychopathology. This was observed in most of the analyses. Notably, once attachment dimensions were accounted for, social support remained a significant predictor of functioning, but with much smaller, and less consistent, effects. Longitudinal research with high risk individuals has indicated that a socially supportive relationship with at least one figure is protective against maladaptation later in life (Egeland, 1997; Werner, 1993). Survivors of child maltreatment have also been reported to benefit from social support (Muller, Sicoli, et al., 2000; Runtz & Schallow, 1997).

Contrary to expectation, the interaction of attachment and social support was not predictive of functioning once the main effects of attachment and social support were accounted for. Attachment, particularly

when characterized as view of self, remained a strong predictor of psychopathology irrespective of its relationship to social support. This result may run contrary to the findings of Ognibene and Collins (1998), where securely attached individuals were more likely to seek social support in response to stress. These researchers did not, however, examine whether or not social support contributed to functioning or psychopathology.

One way of understanding this finding is that those who are securely attached and have a positive view of self may be able to cope adaptively with stress by relying primarily on their inner resources. Research conducted by Mikulincer et al. (see Mikulincer et al., 2003 for a discussion) supports the contention that securely attached individuals possess a sense of self-efficacy, and more often find their inner resources sufficient when dealing with stressors. Similarly, studies suggest that those with attachment styles characterized by a positive self model cope more effectively using their own resources. Those with a positive view of self score higher on measures of optimism, self-confidence, independence, and resourcefulness (Diehl et al., 1998), and are less likely to use escape-avoidance strategies (Ognibene & Collins, 1998). It may be that individuals who are securely attached, and have a positive view of self, carry a greater sense of confidence in their ability to cope effectively with life's stressors, requiring them to depend on external resources (i.e., social support) less frequently.

Types of maltreatment

Psychological abuse was the largest predictor of psychopathology among the abuse variables examined here. Although physical abuse and exposure to domestic violence were significantly correlated with psychopathology, they did not remain clear predictors when considered concurrently with psychological abuse. Due to past difficulties regarding definition and measurement, psychological abuse is just beginning to generate empirical research (Ferguson & Dacey, 1997). Studies have found a high prevalence of psychological abuse, even in normative samples, among both children and adults (Ferguson & Dacey, 1997; McGee, Wolfe, & Wilson, 1997). Moreover, psychological abuse has emerged in other studies as more closely related to psychopathology and adjustment than physical abuse (Ferguson & Dacey, 1997; Higgins & McCabe, 2003; McGee et al., 1997). Researchers suggest that parental hostility and criticism may directly attack the self-system, placing individuals at extreme risk for affective disturbances in adulthood (McGee et al., 1997). Future research needs to delineate more precisely the effects that psychological maltreatment may have on development and adaptation.

Clinical implications

The current study emphasizes the need to address both attachment security and the specific aspect of self in therapy, especially following child maltreatment. Results further suggest that simply having social support available may not be helpful for all individuals. The hypothesis that attachment styles can shift in response to psychotherapy has been supported theoretically and empirically (Bowlby, 1988; Crittenden, 1997; Fonagy et al., 1996). Crittenden (1997) suggests that the therapist can serve as a secure base from which the client can explore threatening experiences, and from which the client can re-work internal models of self and other. Researchers such as Briere (1996) address in treatment difficulties in self-functioning stemming from traumatic experiences.

Limitations

Three areas of concern are worthy of note. First, this study relied solely on paper-and-pencil self-reporting of social support, attachment, psychopathology, and maltreatment. This may have introduced certain biases in responding. Future research would benefit from a multi-method approach to measuring variables of interest, including interviews and behavioral observations.

Second, maltreatment was measured retrospectively, and thus reports may be subject to distorted and/or selective memories. Prospective, longitudinal studies are suggested for future research, following both high and low-risk individuals from childhood to adulthood. Longitudinal research provides more information regarding direction of effects. Thus, this type of research can provide a clearer picture of how protective factors and processes operate over the course of development.

Third, the current sample consisted of undergraduate students; thus, the results may not fully generalize to other populations and age groups. However, the fact that the current results are consistent with those found in studies that employed different populations (Alexander et al., 1998; Mikulincer et al., 1993; Muller & Lemieux, 2000; Muller et al., 2001; Roche et al., 1999) may suggest that these findings are reasonably robust.

Concluding comments

Overall, this study contributes to a growing body of literature emphasizing the importance of attachment security, and particularly view of self, as a protective factor. This research contributes further to understanding of attachment dimensions and their protective effects with regard to both general psychopathology symptoms and trauma symptomatology. While previous research has investigated attachment in smaller samples of individuals with trauma histories, this study found consistent results utilizing a large sample of young adults, both with and without a history of physical maltreatment.

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Résumé

Objectif : Le but de cette étude a été d'examiner le rôle que jouent l'accompagnement et l'attachement dans la psychopathologie de jeunes adultes avec et sans passé de maltraitance physique. Le concept d'attachement a été déterminé par l'évaluation de la perception de soi-même et de celle de l'autre. L'attachement et l'accompagnement ont été étudiés de façon individuelle et de façon simultanée en tant que facteurs de protection.

Méthode : L'échantillon comprenait 956 jeunes adultes, dont 294 avaient un passé de maltraitance physique. Les sujets ont rempli une série de questionnaires concernant l'attachement, l'accompagnement, et les symptômes psychopathologiques. Une étude par régression a été utilisée, pour étudier si l'attachement et/ou l'accompagnement faisaient prévoir une psychopathologie habituelle.

Résultats : Les résultats ont montré qu'un attachement sûr, en particulier s'il est caractérisé par une perception positive de soi-même, faisait prévoir des taux plus bas de psychopathologie, quelque soient les antécédents de maltraitance. Notamment, que la perception de soi-même était un indice prédictif bien plus grand que la perception de l'autre ou qu'un accompagnement chez des sujets avec et sans passé de maltraitance physique. Dans les sous-types de maltraitance, la violence psychologique se révélait comme le meilleur indice prédictif de psychopathologie.

Conclusions : L'étude en cours suggère que la dimension d'attachement et de perception de soi-même est fondamentale pour prédire un niveau de psychopathologie, quelque soit le passé de maltraitance. De plus, la maltraitance psychologique mérite d'autres études en tant que facteur de risque.

Resumen

Objetivo: El propósito de este estudio fue examinar los roles que desempeñan el apoyo social y el juego del apego en relación a la psicopatología en adultos jóvenes con y sin una historia de maltrato físico. El apego fue conceptualizado en terminus de las dimensiones de su vision de su yo y su vision del otro. El apego y el apoyo social fueron examinados individual y concurrentemente como factores de protección.

Método: La muestra consistió en 956 adultos jóvenes, de los que 294 tenían una historia de abuso físico. Los participantes completaron cuestionarios de servicio que preguntaban sobre el apego presente, apoyo social y síntomas psicopatológicos. Se utilizó un diseño de regresión para examinar como el apego y/o el apoyo social predecían la psicopatología actual.

Resultados: Los resultados indicaron que la seguridad del apego, especialmente cuando estaba caracterizada por una vision positiva del yo, predecía fuertemente niveles más bajos de psicopatología, sin importar la condición del abuso. Particularmente, la vision del yo resultó ser un mejor predictor que la visión del otro o el apoyo, para las personas con y sin una historia de maltrato físico. Entre los subtipos de maltrato, el abuso psicológico surgió como el mayor predictor de psicopatología.

Conclusiones: El presente estudio sugiere que la dimension de la vision del yo del apego es fundamental para predecir el nivel presente de psicopatología, sin importar la historia de abuso. Además, el subtipo de abuso psicológico merece más investigación como factor de riesgo.